

Assumptions, diagnostics, and multicollinearity

Last time: Variance inflation factors

Motivation: High multicollinearity means a predictor can be estimated well by some combination of other predictors

Variance inflation factor: Let X_i be a predictor in the model, and R_i^2 the coefficient of determination for a model estimating X_i from the other predictors. The *variance inflation factor* of X_i is

$$VIF_i = \frac{1}{1 - R_i^2}$$

- + Higher VIF means more multicollinearity
- + Rule of thumb: worried when $VIF_i > 5$ (i.e. $R_i^2 > 0.8$)

Class activity

+ $\widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \quad R^2_{adj} = 0.949$

+

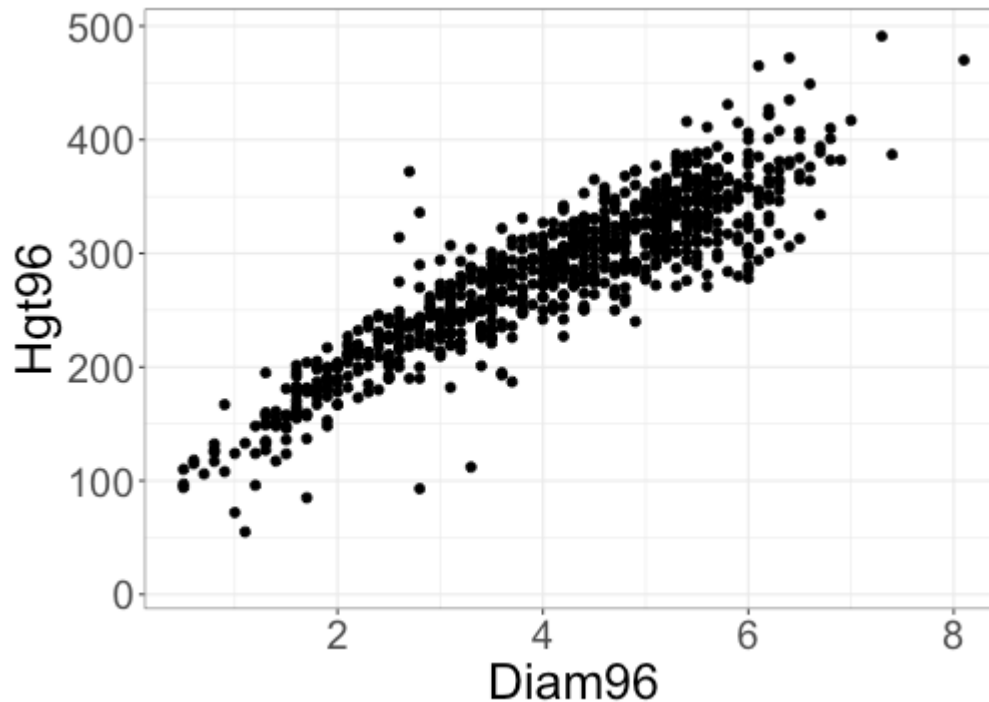
$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96} \quad R^2_{adj} = 0.942$$

Why doesn't R^2_{adj} increase when I add diameter to the model?

Class activity

adding diameter doesn't
help our ability to
predict response
(no new info)

Why doesn't R^2_{adj} increase when I add diameter to the model?



Because diameter and height in 1996 are so correlated.

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

$$\text{VIF} = \frac{1}{1 - R^2}$$

R^2 = coefficient of determination for predicting Diam96 from Hgt96

$$R^2 = 0.813$$

$$\Rightarrow \text{VIF} = \frac{1}{1 - 0.813} = 5.35 > 5$$

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

```
m1 <- lm(Diam96 ~ Hgt96, data = Pines)
summary(m1)$r.squared
```

```
## [1] 0.8134518
```

$$VIF = \frac{1}{1 - 0.813} = 5.35$$

Why is multicollinearity a problem?

Interpretation of individual terms is difficult!

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

Does this mean that tree diameter is negatively associated with height?

no!

- ① usual interpretation: a one-unit in diameter is associated w/ a decrease of 1.26 units in height in 1997, holding height in 1996 fixed
- ② Diam96 & Hgt96 are strongly correlated! Hard to imagine holding height "fixed" and changing diameter

Why is multicollinearity a problem?

Interpretation of individual terms is difficult!

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

Does this mean that tree diameter is negatively associated with height?

No! Model with just diameter:

$$\widehat{\text{Hgt97}} = 161.77 + 46.13 \text{ Diam96}$$

Problem: Diameter and Height in 1996 are strongly correlated; doesn't make sense to fix one and change the other.

Why is multicollinearity a problem?

It is hard to estimate the coefficients!

Suppose $Y = 1 + 2X_1 + 3X_2 + \varepsilon$

and suppose X_1, X_2 are perfectly correlated

e.g., $X_2 = 2X_1$

Then: $(2X_1 + 3X_2) = X_2 + 3X_2 = 4X_2$
 $= (0X_1 + 4X_2)$

$= (8X_1 + 0X_2)$

what are the
true coefficients

$\beta_1, \beta_2?$

There are infinitely many possible values!

Why is multicollinearity a problem?

It is hard to estimate the coefficients!

Model with just height in 1996:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	40.590784	2.524484393	16.07884	5.283829e-51
## Hgt96	1.096059	0.008734402	125.48762	0.000000e+00

Model with both height and diameter in 1996:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	40.244053	3.04659425	13.209522	2.598076e-36
## Hgt96	1.115861	0.02173417	51.341300	2.057358e-260
## Diam96	-1.256481	1.02359067	-1.227523	2.199725e-01

$$t = \frac{\hat{\beta}_1 - 0}{SE\hat{\beta}_1}$$

$SE\hat{\beta}_1$ is much higher!
(harder to estimate = more variability)

harder to reject H_0 if $SE\hat{\beta}_1$ is larger (lose power)

Project data

- + MED_EARNINGS: median earnings for graduates from each university
- + CONTROL: whether the school is public or private
- + SATVRMID: midpoint of SAT critical reading scores of students attending the school
- + SATMTMID: midpoint of SAT math scores
- + SATWRMID: midpoint of SAT writing scores
- + ACTCMMID: midpoint of the ACT cumulative scores
- + ACTENMID: midpoint of ACT English scores
- + ACTMTMID: midpoint of ACT math scores
- + + others...

Which variables might exhibit multicollinearity?

Project data

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- + + others...

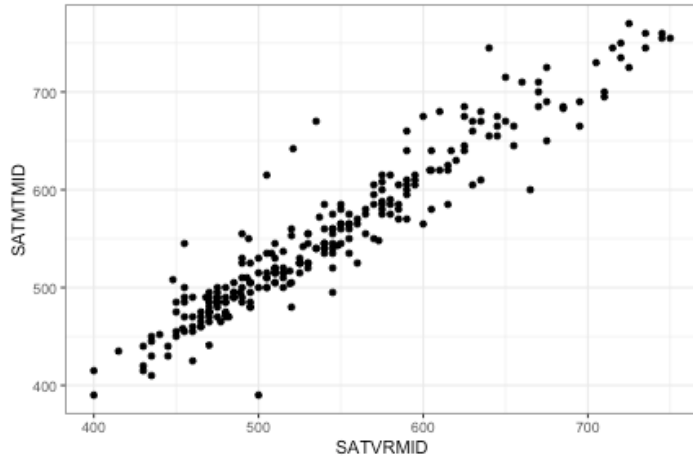
There is probably multicollinearity with all the test score variables

Project data

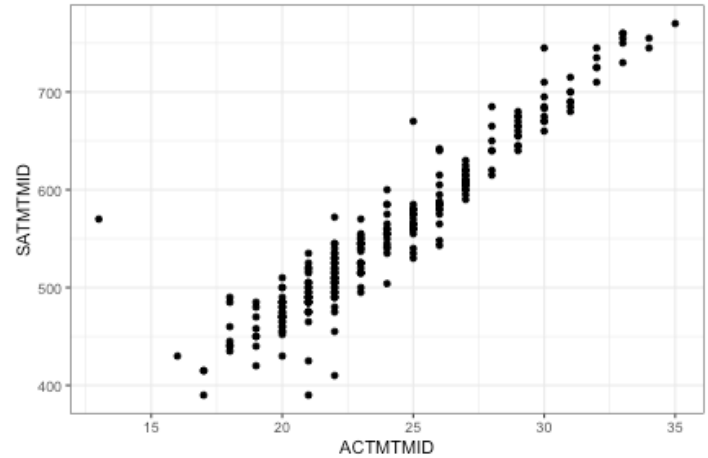
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- + + others...

How can we visually investigate correlated predictors?

Project data



$$r = 0.95$$



$$r = 0.94$$

VIFs in R

Fortunately, R makes it easy to calculate VIFs:

```
library(car)
m1 <- lm(MED_EARNINGS ~ SATVRMID + SATMTMID +
          SATWRMID + ACTCMMID + ACTENMID +
          ACTMTMID + ACTWRMID,
          data = scorecard)
vif(m1)
```

```
## SATVRMID SATMTMID SATWRMID ACTCMMID ACTENMID ACTMTMID AC
## 29.78240 17.69560 27.89062 27.68832 21.90292 17.26265 1
```

- + The `vif` function (from the `car` package) calculates a VIF for each predictor
- + The variance inflation factor for SAT verbal score is 29.78, e.g.

Handling multicollinearity

We have a few options for handling multicollinearity:

- + Remove variables
 - + Example: just pick one of the test score variables, and discard the rest, since they are so highly correlated

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 - + Example: take an average or sum of the test score variables, instead of using all of them individually

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- + Create a new variable
 - + Example: take an average or sum of the test score variables, instead of using all of them individually
- + Ignore it
 - + If we don't care about interpreting coefficients or doing inference, multicollinearity doesn't matter as much

Class activity

Work on the class activity (in R).

$$y = 1 + x_1 + x_2 + \varepsilon$$

$$\beta_1 = 1$$

Class activity

```
rho <- 0
X <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = matrix(c(1, rho, rho, 1), 2, 2))
x1 <- X[,1]
x2 <- X[,2]

y <- 1 + x1 + x2 + rnorm(n)
m1 <- lm(y ~ x1)
m2 <- lm(y ~ x1 + x2)
coef(m1)
```

```
## (Intercept)          x1
##      1.124596      1.073064
```

```
coef(m2)
```

↑ similar, b/c adding x_2
↓ doesn't change anything
about x_1

```
## (Intercept)          x1          x2
##      1.1129517      1.0202536      0.9645505
```

Class activity

```
rho <- 0.9
X <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = matrix(c(1, rho, rho, 1), 2, 2))
x1 <- X[,1]
x2 <- X[,2]

y <- 1 + x1 + x2 + rnorm(n)
m1 <- lm(y ~ x1)
m2 <- lm(y ~ x1 + x2)
coef(m1)
```

$$y = 1 + x_1 + x_2 + \varepsilon$$

if $x_1 \approx x_2$ (high correlation)

$$y \approx 1 + 2x_1 + \varepsilon$$

$$1 + 2x_2 + \varepsilon$$

```
## (Intercept)          x1
## 0.9361277      1.8706363
```

$\hat{\beta}_1 \approx 2$

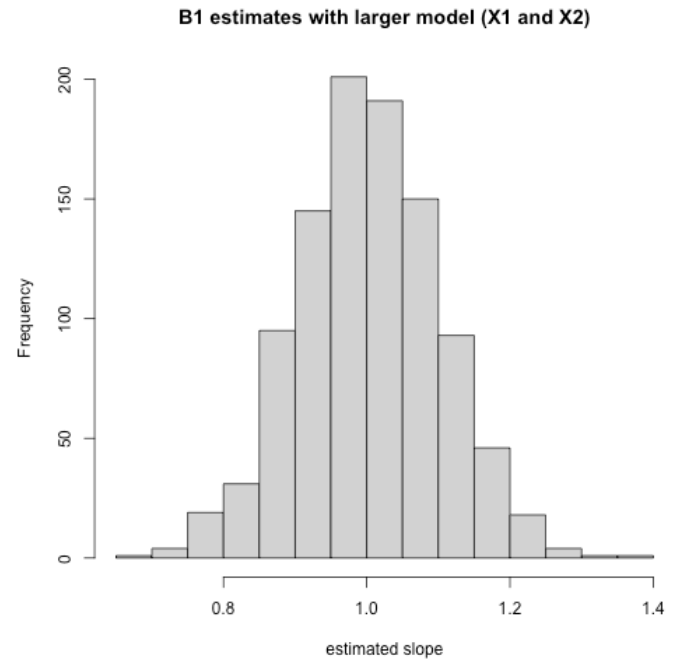
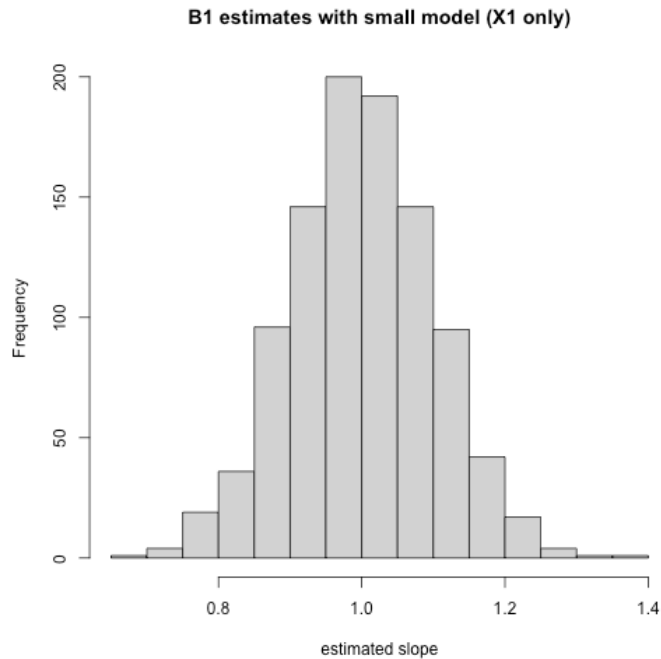
```
coef(m2)
```

very different

```
## (Intercept)          x1          x2
## 0.9446137      1.3926972      0.5789134
```

Class activity

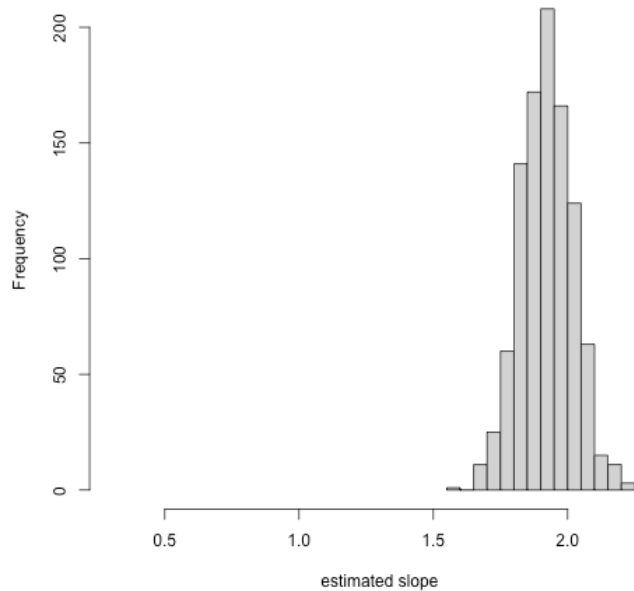
When correlation = 0:



Class activity

When correlation = 0.9:

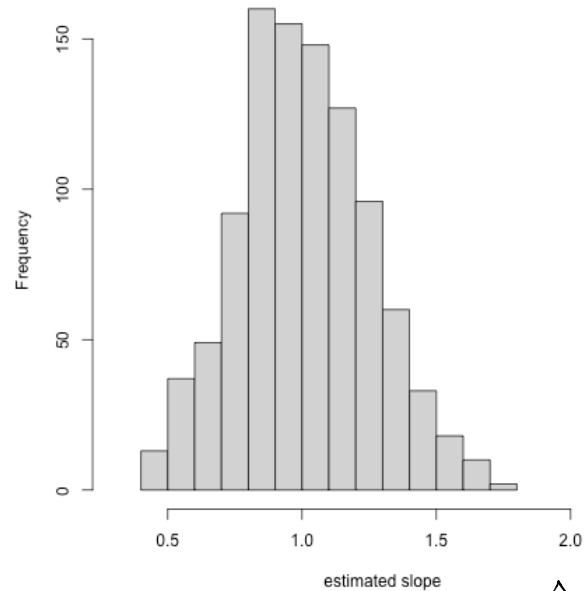
B1 estimates with small model (X1 only)



$$\text{variance}(\hat{\beta}_1) \approx 0.009$$

$$\frac{0.045}{0.009} \approx 5$$

B1 estimates with larger model (X1 and X2)



$$\text{variance}(\hat{\beta}_1) \approx 0.045$$

$$\approx \frac{1}{1 - 0.9^2} \quad (\text{VIF})$$